



Information flow redistribution for diabetic retinopathy multi-lesion segmentation[☆]

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HIGHLIGHTS

- Identify causes of false positives/negatives in deep neural networks for lesion segmentation.
- Maintain multi-level feature richness to enhance lesion detection accuracy.
- Introduce cross-spatial/channel information flows (CSIF/CCIF) to redistribute features.
- Design window-guided local feature reassignment for precise lesion details.

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ABSTRACT

Common retinal lesion segmentation methods often rely on multi-stage networks, which cause spatial information to diminish in deeper layers while channel information increases, leading to feature imbalance. To address this, we propose a novel and effective method that introduces a top-down Cross Spatial Information Flow (CSIF) and a bottom-up Cross Channel Information Flow (CCIF). CSIF gradually propagates fine spatial details from shallow to deep layers, while CCIF delivers precise channel information from deep to shallow layers. Specifically, we extract fine spatial cues and accurate channel features via autocorrelation-based affinity operations in spatial and channel dimensions, respectively. These are then redistributed using cross-correlation mechanisms to compensate for insufficient spatial or channel information at various depths, while irrelevant features are compressed to maintain computational efficiency. We apply this method to the engineering task of retinal lesion segmentation, and experimental results on two widely used public datasets demonstrate that our method achieves competitive performance and strong generalization capabilities across different data domains.

1. Introduction

The challenge of medical image segmentation lies in accurately delineating the detailed features of the region of interest, which contains numerous implicit pathological information that can significantly influence diagnostic outcomes [1]. Therefore, early screening and timely intervention are crucial for preventing disease progression and alleviating patient suffering [2]. Through the automation of the diagnostic process, not only can the identification of diabetic retinopathy be accelerated, but the likelihood of human error can also be reduced [3]. Retinal fundus image abnormalities include red lesions such as microaneurysms (MA), hemorrhages (HE), and bright yellow lesions such as hard exudates (EX) and soft exudates (SE) [4].

Taking Fig. 1 as an example, retinal fundus images often contain numerous subtle lesion features, with various lesion types clustering together. EX and SE lesions, both appearing as bright yellow spots, are difficult to distinguish. HE lesions are similar in color to retinal blood vessels, while MA lesions are extremely small and nearly imperceptible. As shown in the left part of the figure, these lesions exhibit high similarity to the background, making them extremely difficult to separate effectively. Consequently, during retinal lesion segmentation, such lesion information is easily overlooked.

To tackle these challenges, it is essential to design models capable of preserving both spatial detail and discriminative channel representation across network layers. Conventional encoder-decoder and

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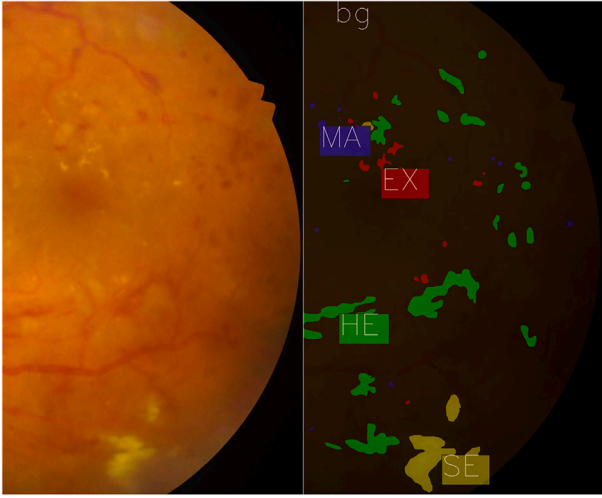


Fig. 1. Visualization of the retinal disease segmentation task: yellow denotes soft exudates (SE), green indicates hemorrhages (HE), red represents hard exudates (EX), and blue corresponds to microaneurysms (MA).

attention-based methods often fail to maintain this balance, resulting in incomplete activation of subtle lesion regions. Our work addresses this problem by introducing a dual-directional feature interaction framework that explicitly reassigns information across layers through two complementary mechanisms: Cross Spatial Information Flow (CSIF) and Cross Channel Information Flow (CCIF).

These lesion features are prone to being overlooked during retinal lesion segmentation. Such omissions can lead to severe misdiagnosis or missed diagnoses, causing patients to lose the best opportunity for treatment. These conditions impair vision and, if left untreated, may lead to partial blindness [5]. As a result, this task is typically handled by highly skilled and experienced doctors. However, manual segmentation is a time-consuming and labor-intensive process, making it challenging to achieve widespread implementation in regions with limited medical resources. To alleviate healthcare professionals from this intensive workload, significant efforts have been made to utilize deep learning methods for the automated segmentation of regions of interest in images. These efforts have greatly mitigated the problem of human resource shortages. These studies primarily introduce attention mechanisms to enhance the model's focus on key regions and emphasize feature enhancement operations. However, they overlook the issue of imbalanced spatial and channel information distribution among features extracted by learning factors at different layers. Addressing this research gap, this study proposes a novel cross-layer information flow redistribution mechanism, inspired by cross-layer information modeling. The method is designed from the perspective of maintaining feature representation integrity.

Specifically, our framework establishes top-down and bottom-up information pathways that facilitate bidirectional communication between shallow and deep features. CSIF progressively propagates fine spatial details from shallow layers to deep layers, preserving lesion localization accuracy, whereas CCIF transfers discriminative channel semantics from deep layers to shallow layers, enhancing feature differentiation. This dual information flow design ensures that each feature layer retains a sufficient balance of spatial and semantic cues, which is crucial for precise lesion boundary recovery and small-lesion detection.

However, the diverse appearance of lesions and complex background interference significantly reduce the performance of pixel-level predictions [6]. To ensure the accurate extraction of detailed features, researchers have begun to pursue higher-resolution representations. However, there exists a significant gap between medical images and natural images, making it difficult to achieve good results by directly applying these methods. This is due to several reasons:

1. **Size disparity in lesions:** In medical images, certain lesions have extremely small sizes, resulting in sparse pathological distributions. Conversely, other lesions exhibit significantly larger size distributions. This substantial size disparity poses challenges for networks to adapt effectively. Intuitively, small lesion regions require networks to extract as much detailed information as possible within a small receptive field, while also demanding a high sensitivity to subtle feature variations. In contrast, large lesion regions necessitate a large receptive field for capturing long-range contextual dependencies.
2. **Blurred boundaries and low pixel variance:** Medical images often exhibit blurred boundaries between structures and relatively low pixel variance. This necessitates more precise operations on the information embedded in medical images to ensure accurate analysis.
3. **Clustered lesions:** Different types of lesions frequently cluster together, making representation learning even more challenging due to the need to distinguish closely packed and similar features.

One important reason for these problems is the imbalance in the distribution of spatial and channel information among features extracted by different layers during feature learning. However, most networks directly recombine features after applying learning operators, relying solely on learned weights for sampling. This approach makes it difficult to effectively integrate spatial and channel information during learning, often leading to the loss of important regions with low activation responses.

Motivation: As the network deepens, the spatial scale of features diminishes while the channel scale increases. During this process, the spatial information implicitly contained in features gradually decreases while channel information gradually increases, meaning shallow features possess more refined spatial characteristics while deep features have more accurate channel characteristics. The imbalance in information distribution across different feature layers tends to worsen progressively, making it increasingly challenging for models to identify critical features.

To address this issue, we propose the methods illustrated in Fig. 2. Through multi-level feature interaction, the network can simultaneously extract both fine and sparse lesions, as well as lesions with larger size distributions, across the multi-scale receptive fields of different layers. Starting with the accurate channel information in bottom-level features and the fine spatial information in top-level features, we utilize an affinity operation to extract spatial texture features and channel semantic features. These features are then redistributed in an information flow manner to enhance deep features lacking spatial information and shallow features lacking channel information before feature recombination. Sufficient feature information enables the network to better distinguish the blurred boundaries in medical images, allowing for more precise operations. Furthermore, window-based regional operations are introduced to precisely locate potential lesions, overcoming the challenges posed by clustered lesions of different types.

Contributions:

1. We design a top-down cross-spatial information flow method and a bottom-up cross-channel information flow method. These methods redistribute the spatial information of shallow features and the channel information of deep features via information flow, ensuring that all features possess sufficient information to activate lesions.
2. To achieve optimal redistribution for each feature layer, we develop a spatial information redistribution module and a channel information redistribution module. These modules leverage affinity operations and cross-distribution mechanisms at the spatial and channel levels to allocate enriched spatial and channel information to all feature layers.

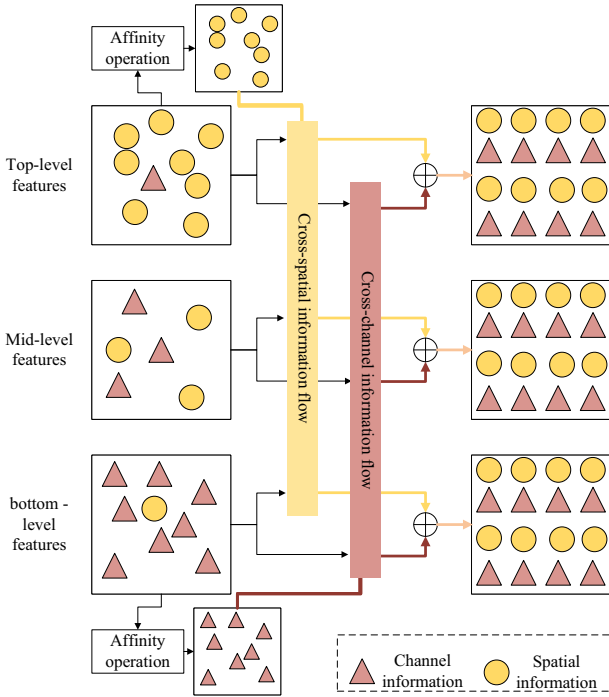


Fig. 2. Information flow redistribution: spatial information is extracted from top-level features, channel information is derived from bottom-level features, and cross-spatial information flow (CSIF) and cross-channel information flow (CCIF) are leveraged to redistribute them.

3. Comprehensive experiments on publicly available datasets demonstrate that our proposed method achieves competitive performance in multi-lesion segmentation tasks. Additionally, the generalization study shows that the proposed method exhibits strong generalization capabilities.

2. Related work

2.1. Attention mechanisms

In recent years, semantic segmentation has made significant progress, with one major direction focusing on enhancing high-resolution features through feature enhancement mechanisms. Typically, attention mechanisms enhance feature representations in semantic segmentation networks by aggregating contextual information. To better explore spatial dependencies, [7,8] design modules such as non-local networks and disentangled non-local neural networks (DNL) to aggregate pixel-level contextual information through spatial dependencies and attention maps. Chen et al. [9] aggregate context information based on pixel positions using a mild weighting mechanism. Zhao et al. [10] employ pyramid scene parsing methods to gather global context information based on different areas. Zhao et al. [11] achieve the aggregation of long-distance contextual information by interconnecting attention maps from each position. He et al. [12] use global image representations to compute affinities among different areas, thereby aggregating multi-scale contextual information. Huang et al. [13] apply a cross-striped attention mechanism to aggregate global contextual information on a pixel-by-pixel basis. Huang et al. [14] employ a ReMixed Transformer Context Bridge to capture global correlation and local contextual information from multi-scale features. Liu et al. [15] use the Multi-level Feature Aggregation (MFA) module to adaptively compute the contextual weights of different feature layers.

Another approach is to significantly enhance features at specific locations to ensure that the network can better focus on foreground information. To better capture relevant features, [16] utilizes the RPEC

attention mechanism to capture the primary information within the features. Zhang et al. [17] highlight the significance of specific regions across layers using cross-layer feature enhancement (CLFE). Xu et al. [18] employ fusion modules to adaptively suppress background noise while accentuating foreground-related information. Wang et al. [19] design a Deformable Attention Convolution (DAC) module to distinguish foreground information through spatial distribution. Chen et al. [1] employ the coordinated interaction between spatial attention and channel attention components to address noise interference issues. The CSCA U-Net [20] proposes a Channel and Spatial Composite Attention (CSCA) mechanism to enhance feature representation by increasing network depth and incorporating Dual Squeeze-and-Excitation (DSE) blocks in the bottleneck layer. The PGKD-Net [21] improves feature learning through Multi-scale Context Aggregation (MSCA) and Multi-level Feature Fusion (MLFF). Liu et al. [22] leverage a spatial pyramid module and an attention mechanism to further refine feature extraction and representation.

In recent years, several Transformer-based networks have been successfully evaluated in medical image segmentation, achieving strong global modeling capabilities. TransUNet [23] was the first attempt to merge Transformers with U-Net, demonstrating the feasibility of a hybrid architecture. It incorporated a Transformer pathway after the originally convolution-based U-Net encoder to model long-range dependencies further. TransFuse [24], in a parallel fashion, extracts context and long-range dependencies through both a CNN encoder and a Transformer encoder. It subsequently combines CNN and Transformer features through a CNN decoder followed by a second CNN decoding step. DS-TransUNet [25] incorporates a hierarchical Swin Transformer dual-scale encoder for multi-scale feature learning, enhancing the model's flexibility and capabilities.

These methods have not only achieved exceptional success in natural scene image segmentation but have also paved the way for lesion segmentation in fundus images. However, these approaches often limit their scope to augmenting contextual information, overlooking the substantial spatial and channel information distribution imbalance across different feature layers. This imbalance can hinder the learning operators from gaining a comprehensive understanding of the feature information. While these methods bring significant accuracy improvements, they typically facilitate spatial or channel interactions only at selective layers or via post-hoc fusion. In contrast, our proposed CSIF + CCIF scheme explicitly redistributes spatial and channel information across every layer. This ensures each feature map receives both shallow spatial priors and deep discriminative channel cues, allowing more robust activation of subtle lesion regions under constrained computation.

2.2. Lesion segmentation

Previous methods based on convolutional neural networks have approached the multi-lesion segmentation task by decomposing it into multiple independent single-lesion segmentation tasks. These methods trained four separate network models for four types of lesions (SE, EX, MA, HE) [26]. Although this method appears to yield good results, this computationally expensive approach is not suitable for practical medical diagnostics.

Recent lesion segmentation solutions advocate treating multiple lesion segmentation tasks as multi-label tasks, implementing the simultaneous segmentation of four types of lesions within a single model. Zhang et al. [27] attempted to frame the multi-lesion segmentation task as a multi-view task and used two hierarchical models, U-Net and TransU-Net, to leverage the contextual information to guide the model's attention. Pavani et al. [28] employed two parallel encoders to extract texture features. Qiu et al. [29] introduced a Shared Feature Extraction Module (SFEM) between the two parallel encoders to assist the dual-stream learning framework in better segmenting small lesions. Huang et al. [30] proposed that certain lesions may have specific correlations

with particular blood vessels and attempted to integrate the relationships between vessels and lesions through cross-attention mechanisms. Guo et al. [31] utilized multi-scale fusion methods to address the issue of small lesions. Liu et al. [32] rethought the feature reorganization operator and proposed a new upsampling and downsampling method to maintain the activation of small lesions and capture long-range dependencies as much as possible.

Monemian and Rabbani [33] proposed a novel red-lesion detection method in Scientific Reports, which leverages unique morphological features of retinal lesions to achieve robust and high-precision detection. Santos et al. [34] introduced an end-to-end detection framework based on a YOLO deep neural network architecture, significantly improving the efficiency of retinal lesion localization. Furthermore, Sebastian et al. [35] provided a comprehensive survey, summarizing recent advances in diabetic retinopathy lesion detection and segmentation, and emphasizing that the integration of multi-scale feature representation and contextual awareness is a key direction for future research.

In contrast to these methods, our work focuses on the completeness of feature block information. By adopting an information flow perspective and utilizing affinity between features, we ensure that each feature layer contains more complete spatial and channel information, thus improving the network's ability to recognize complex structures.

To achieve this, we propose a method that extracts spatial information from shallow features and progressively distributes it to deeper layers, while extracting channel information from deep features and gradually allocating it to shallow layers. This approach ensures that each feature block has sufficient spatial and channel information, providing ample support for subsequent feature extraction and effectively improving segmentation performance.

Recent studies from other engineering domains have also highlighted that data-driven neural models often struggle when the available information is incomplete, imbalanced, or degraded. For example, neural estimators become highly sensitive to noise and insufficient feature representation, requiring additional structural constraints to ensure stable convergence [47]. Similarly, in navigation research, advanced recurrent models have been used to compensate for missing or disrupted measurements by reconstructing long-term dependencies in the input data [48]. These observations are consistent with our motivation: medical images also contain inherently unbalanced spatial and channel information across layers, and effective feature redistribution is essential for robust segmentation.

3. The proposed methods

To address the imbalance between spatial and channel information in deep feature representations, this section presents the proposed Information Flow Redistribution (IFR) framework. The design rationale,

network architecture, and detailed mechanisms of the cross-spatial and cross-channel flows are described in the following subsections.

As the network depth increases, the spatial scale of features continues to shrink while the channel scale deepens, meaning that shallow features lack more accurate channel characteristics, and deep features lack finer spatial characteristics. Some methods adopt strategies to complement the missing information in subsequent steps; however, this does not imply that the absence of this information will not adversely affect the current feature extraction process. On the contrary, we believe that this information deficiency will lead to an inability to achieve comprehensive modeling of the features during the feature extraction process, and such deficiencies will gradually accumulate, significantly impacting the final results.

To better address this situation, we propose an information flow redistribution (IFR) as shown in Fig. 3, which is primarily composed of CSIF (Top-down Cross-Spatial Information Flow), W-SIR (Windowed Spatial Information Redistribution), W-CIR (Windowed Channel Information Redistribution), and CCIF (Bottom-up Cross-Channel Information Flow). Specifically, our model operates in multiple stages: Stage 1: A high-resolution feature is extracted directly from the input image using high-resolution convolutional blocks. This feature inherits minimal spatial information imbalance as it directly originates from the input image. Stage 2–4: As high-resolution convolutions extract multi-level high-resolution features from different feature layers, spatial and channel information imbalances emerge across these layers.

To achieve this, we propose a Top-Down Cross-Spatial Information Flow (CSIF) mechanism, which channels fine spatial information from shallow features to deep features. This improves the spatial representation capability of deep features that often lack such information. Similarly, we introduce a Bottom-Up Cross-Channel Information Flow (CCIF) mechanism to flow accurate channel information from deep features to shallow features, enhancing the channel representation capability of shallow features. By redistributing spatial and channel information, the features become more enriched, enabling the network to better distinguish blurred boundaries in medical images and perform more precise operations.

Algorithm 1 outlines the overall IFR process. It integrates Top-Down Cross-Spatial Information Flow (CSIF) and Bottom-Up Cross-Channel Information Flow (CCIF) in a unified bidirectional manner, ensuring that each layer receives both fine-grained spatial priors and discriminative channel semantics.

3.1. Top-down cross-spatial information flow (CSIF)

This subsection introduces the CSIF mechanism, which transfers fine-grained spatial information from shallow to deep layers to enhance spatial precision and boundary delineation in deeper features. Repeated downsampling operations inevitably degrade the spatial characteristics

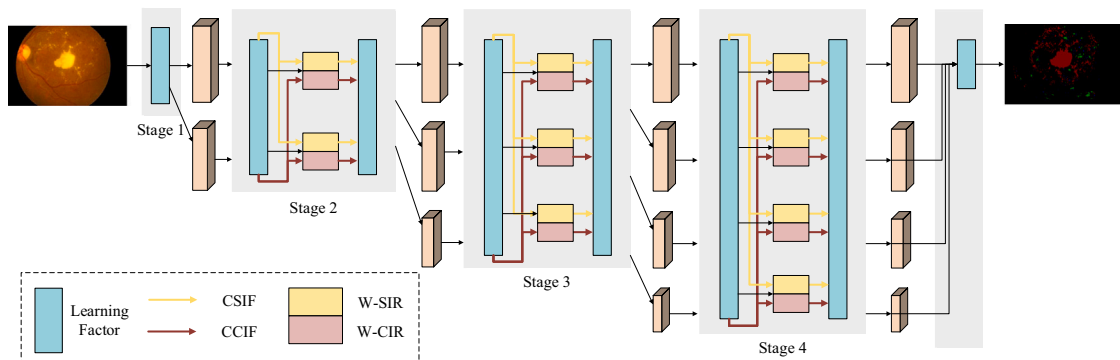


Fig. 3. Overview of our IFR (Information Flow Redistribution). The proposed CSIF and W-SIR modules are used for spatial information redistribution, while CCIF and W-CIR modules are used for channel information redistribution.

Algorithm 1 Information flow redistribution (IFR) framework.

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1: Set  $h_f \leftarrow x_1$   $\triangleright$  High-resolution spatial reference (shallowest layer)
2: Set  $d_f \leftarrow x_m$   $\triangleright$  Deep semantic channel reference (deepest layer)
3: for each layer  $i = 1$  to  $m$  do
4:    $f_{sp} \leftarrow \text{CSIF\_W-SIR}(h_f, x_i)$ 
5:    $f_{ch} \leftarrow \text{CCIF\_W-CIR}(x_i, d_f)$ 
6:    $y_i \leftarrow f_{sp} + f_{ch} + x_i$   $\triangleright$  Residual fusion
7: end for
8: return  $\mathcal{Y}$ 

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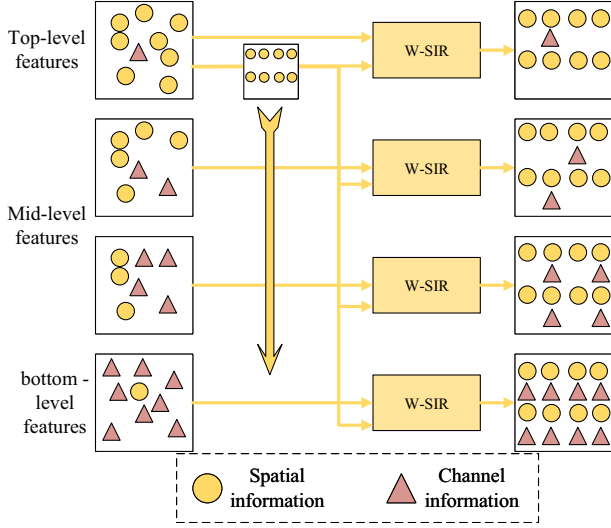


Fig. 4. Top-down cross spatial flow (CSIF).

of deep features, causing difficulties for subsequent learning operators, as they fail to maintain a comprehensive understanding of feature information. Considering that shallow features have more detailed spatial information, we propose a top-down cross-spatial flow method, as illustrated in Fig. 4, whose main function is to redistribute the spatial information of multiple features to ensure that both deep and shallow features can better emphasize the important spatial positions in the feature maps, preventing the loss of critical information during the feature reorganization process.

Specifically, in each stage of IFR, we continue to use high-resolution convolutions to extract the feature information $X = \{x_1, x_2, \dots, x_i\}$ of the image at different levels, and then the features X are processed sequentially from top to bottom according to the following formula to redistribute spatial information to each feature block. W-SIR calculates the affinity between the target input and the reference input, redistributing spatial information accordingly:

$$\hat{x}_i = W_{\text{SIR}}(x_1, x_i), \quad i \in \{1, 2, \dots, m\}, \quad (1)$$

Here, x_1 represents the top-level feature, and m denotes the number of input features. W_{SIR} refers to the *Windowed Spatial Information Redistribution*, which will be explained in detail in the following sections. To facilitate understanding, the step-by-step process of the CSIF mechanism is described below.

- Reference selection:** The shallowest high-resolution feature x_1 is selected as the spatial reference because it preserves the most fine-grained positional details from the input image.
- Projection and alignment:** The learnable projections ϕ_k and ϕ_v are applied to the reference x_1 , while ϕ_q is applied to the target feature x_i . The reference is spatially aligned with the target feature

through strided operations, yielding the matrices K , V , and Q of dimensions $(C' \times H_i \times W_i)$.

- Window partitioning and affinity computation:** The spatial domain is divided into n windows, each containing $h \times w$ positions. For each relative position p within a window, the corresponding vectors are stacked to form Q_p , K_p , and V_p . The similarity is computed as $S_p = Q_p K_p^T / \sqrt{C'}$, and the affinity map is obtained through normalization as $A_p = \text{Softmax}(S_p)$.
- Aggregation and refinement:** The aggregated representation H is derived based on A_p , followed by Layer Normalization and a Feed-Forward Network (FFN) to enhance representational capacity.
- Reprojection and fusion:** The refined representation H is projected back to the target channel dimension and fused with the original target feature through gated element-wise multiplication and residual connection.

At different layers, we take the corresponding layer features x_i ($i \in \{1, 2, \dots, m\}$) as target inputs, while the reference input is always fixed as the shallowest layer feature x_1 . In particular, for the shallowest layer feature x_1 , since it already represents the most comprehensive spatial information at the current stage, we use this feature as both the reference and the target feature. Through this information flow mechanism, spatial information from shallow features is progressively transferred to deep features. This redistribution ensures that all feature layers are enriched with sufficient spatial information to activate lesion-related features effectively. Such enhancement enables the network to better address the challenges of medical image segmentation, particularly in detecting and delineating subtle lesions.

3.2. Windowed spatial information redistribution (W-SIR)

The W-SIR module is designed to efficiently redistribute spatial information at the window level. It ensures localized spatial alignment between shallow and deep features while minimizing computational overhead. To achieve the allocation of feature space information for different layers relying on reference input under the constraint of limited computational resources, we extract windowed spatial information from reference features and assign it to target features in a learnable manner. This module divides both reference features and target features into n feature blocks, ensuring that the number of blocks is the same. By performing cross-comparison of spatial features from blocks at different positions in the reference and target features, we gather contextual information and further enhance the feature space information using this context.

As shown in Fig. 5, given the reference input $X \in \mathbb{R}^{C_x \times H_x \times W_x}$ and the target input $Y \in \mathbb{R}^{C_y \times H_y \times W_y}$, the CSIF mechanism ensures that each layer retains sufficient spatial priors by employing the W-SIR module to extract fine-grained spatial cues from the shallow high-resolution reference and inject them into deeper target layers. The module first utilizes a convolution block with batch normalization to compress the channel features of X and Y into a 1-dimensional representation, focusing the features on expressing spatial information. During this process, the convolution block C-conv employs a 1×1 convolution, while the convolution block C-convd uses a strided convolution to transform the spatial dimensions of the reference input to correspond with the target input dimensions, denoted as $Y_q, X_k, X_v \in \mathbb{R}^{1 \times H \times W}$.

$$\begin{aligned}
 Y_q &= \phi_q(Y) \in \mathbb{R}^{1 \times H \times W}, \\
 X_k &= \phi_k(X) \in \mathbb{R}^{1 \times H \times W}, \\
 X_v &= \phi_v(X) \in \mathbb{R}^{1 \times H \times W}.
 \end{aligned} \quad (2)$$

Here, ϕ denotes a learnable projection, and H and W represent the width and height of the target input feature. In this process, the model focuses primarily on the spatial information within the features and integrates the critical spatial cues across different feature channels.

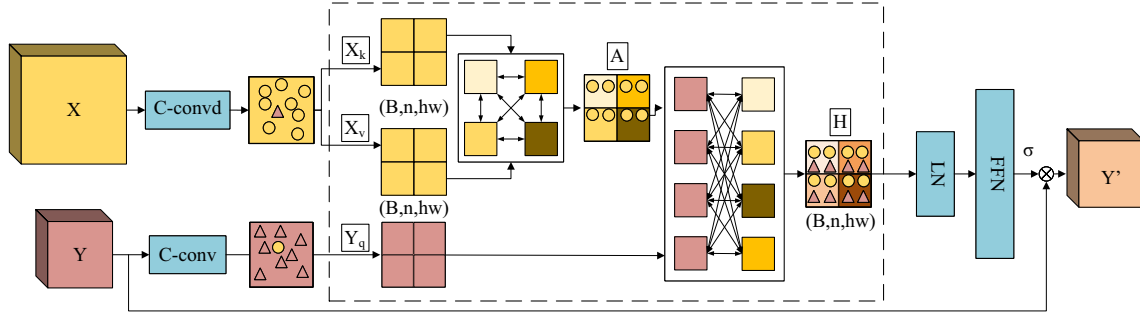


Fig. 5. Windowed spatial information redistribution module (W-SIR).

Next, spatial relative position aggregation divides each feature map into n windows (each containing $h \times w$ positions, where $H \times W = n \times h \times w$). For any relative position p within a window, the corresponding vectors are stacked to form $K_p = X'_k[:, p] \in \mathbb{R}^n$ and $V_p = X'_v[:, p] \in \mathbb{R}^n$, where X'_k and $X'_v \in \mathbb{R}^{n \times hw}$, n represents the number of blocks, and hw denotes the window size. Considering that X'_k and X'_v contain fine spatial features inferred from shallow layer features, we can obtain an affinity map A of size (hw, hw) through an affinity operation. When calculating the affinity map at a given position, all other features within that position's window are referenced and compared with the features at the same relative position in other windows. Therefore, the relevant operation at that position can be defined as follows:

$$a_l = x_l^k (x_l^v)^T, \quad l \in \{1, \dots, hw\}, \quad (3)$$

where a_l represents the correlation between the relative position feature and the overall feature. A window-wise softmax operation is then applied over the spatial dimensions to aggregate contextual information and obtain the desired windowed spatial affinity A .

Subsequently, measures are taken to better assign spatial information to the target features. Here, we apply the same windowing operation to transform Y_q into $Y'_q \in \mathbb{R}^{n \times hw}$. At each relative position l in Y'_q , we can obtain a vector set $\Omega_l \in \mathbb{R}^{n \times b}$, which consists of all features in Y'_q that share the same relative position l across different windows. The process of generating windowed cross-spatial information can be defined as:

$$h_l = \sum_{i=1}^{n \times n} a_{l,i} \Omega_{l,i}, \quad l \in \{1, \dots, hw\}, \quad (4)$$

where h_l represents the feature vector of the spatial information at the relative position l . We apply LayerNorm to h to obtain the final desired cross-spatial information H . In this process, both the contextual information within the window and the contextual information across windows are added to H to enhance the representation of spatial information.

Benefiting from the early compression of irrelevant channel information, our structure efficiently aggregates spatial contextual information with minimal computational overhead. It further enhances spatial information directionally within specific windows through interactions between different windows. Subsequently, we apply a Feed-Forward Network (FFN) to enable H to gain nonlinear learning capabilities. Finally, we use the Hadamard product on the target feature Y and the windowed cross-spatial information H , resulting in a spatially redistributed feature Y' .

Algorithm 2 outlines the overall CSIF process. The CSIF module transfers fine spatial cues from shallow layers to deeper layers through windowed spatial information redistribution (W-SIR). By learning local affinities within and across windows, it enhances the spatial representation of deep features while maintaining computational efficiency.

Algorithm 2 Top-down cross-spatial information flow (CSIF) with W-SIR.

- 1: Align reference: $h' \leftarrow \text{ConvStride}(h_f, \text{stride} = 2^l)$
- 2: Channel compression: $q \leftarrow \text{Conv1x1}(x_i)$, $k, v \leftarrow \text{Conv1x1}(h')$
- 3: Partition q, k, v into n windows of size $(h \times w)$
- 4: **for** each window $\omega = 1$ to n **do**
- 5: Compute affinity: $A_\omega = \text{Softmax}(q_\omega k_\omega^T / \sqrt{d})$
- 6: Aggregate spatial context: $h_\omega = A_\omega v_\omega$
- 7: **end for**
- 8: Integrate all windows: $H \leftarrow \text{LayerNorm}(\text{FFN}(\{h_\omega\}))$
- 9: Spatial gating map: $M_{sp} = \sigma(H)$
- 10: Apply spatial modulation: $y_{sp} = x_i \odot M_{sp}$
- 11: **return** y_{sp}

At each stage, the W-SIR module is applied progressively from top to bottom to inject spatial information into every feature layer, ensuring that effective spatial cues are stably and efficiently utilized across layers. In this process, the shallow features are first compressed and normalized to eliminate redundant channel interference, thereby maintaining spatial comparability among different layers. These processed shallow features are then divided into multiple local regions and aligned with the deep features at corresponding spatial positions. By computing the similarity between local regions, the network identifies the most representative fine-grained spatial details within the shallow features and aggregates them before injecting them into the deeper layers. To avoid introducing noise, irrelevant or weakly correlated regions are automatically suppressed during the aggregation process. As a result, the enhanced deep features retain their original semantic abstraction capability while acquiring additional fine-grained spatial localization signals from the shallow layers. This mechanism ensures that the output of each layer contains sufficiently complete spatial information, enabling the network to achieve higher spatial sensitivity when detecting and segmenting lesion regions.

3.3. Bottom-up cross-channel information flow (CCIF)

The CCIF mechanism complements CSIF by propagating discriminative channel information from deep to shallow layers. This process strengthens the semantic representation and enhances feature discrimination at early network stages. In contrast to deep features, shallow features are limited by the number of channel dimensions, making it hard to achieve better accuracy. Similarly, directly reorganizing these features would make it difficult for the feature reorganization operator to accurately grasp important channel information. Considering that deep features with a large number of channels often possess better channel information expression capabilities, we designed a bottom-up cross-channel information flow method, as shown in Fig. 6. We hope that through channel information flow, the precise channel information

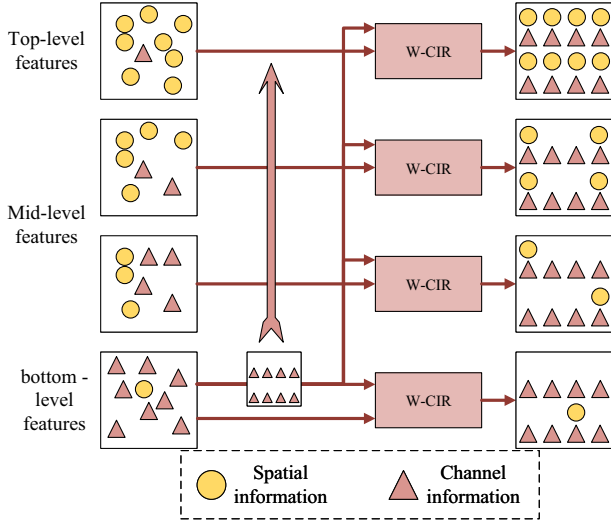


Fig. 6. Bottom-up cross-channel flow (CCIF).

contained in deep features can be mapped to shallow features, allowing the model to accurately discern important channel distributions.

The bottom-up cross-channel information flow adopts a parallel method similar to that of the top-down cross-spatial information flow. After using high-resolution convolutions to extract the feature information $X = \{x_1, x_2, \dots, x_m\}$ from different levels of the image, to further enhance feature redistribution, X is sequentially processed in a bottom-up manner using the Windowed Channel Information Redistribution (W-CIR) module, redistributing channel information to each feature block layer by layer:

$$x_i'' = \text{W-CIR}(x_m, x_i), \quad i \in \{1, 2, \dots, m\}. \quad (5)$$

Here, x_m represents the bottom-level features, and m denotes the number of input features. W-CIR refers to the Windowed Channel Information Redistribution, which will be elaborated on in the following section. The implementation concept of CCIF is analogous to that of CSIF, with the primary distinction being that CSIF operates along the spatial dimension, whereas CCIF performs computations along the channel dimension. Through this strategy, the rich semantic representations in the deep-layer channels guide and enhance the discriminative capability of the shallow features, enabling more precise and context-aware feature refinement across the network hierarchy.

In this structure, we similarly take the features x_i ($i \in \{1, 2, \dots, m\}$) from different layers as target inputs. Unlike the top-down cross-spatial information flow, in this phase, the reference input is always fixed as the deepest layer feature x_m . Similarly, for the deepest layer feature x_m , given that it already represents the most accurate channel information

at the current stage, when allocating channel information for this layer, we use this feature as both the reference and the target feature. Through this information flow mechanism, the channel information from deep features is progressively transferred to shallow features, ensuring that all feature layers are enriched with sufficient channel information to activate lesion-related features effectively.

3.4. Windowed channel information redistribution (W-CIR)

W-CIR enables adaptive channel reallocation within and across feature windows, refining channel weights and improving the consistency of lesion activation across layers. To ensure that the model can effectively allocate information among different layer feature channels based on reference inputs, we extract windowed channel information from the reference features and assign it to the target features in a learnable manner. This module divides both reference and target features into the same number of n feature blocks, cross-referencing the channel characteristics between feature blocks in different locations of the reference and target features to collect contextual information, which is then utilized to enhance the feature channel information.

As shown in Fig. 7, after obtaining the reference features X and the target features Y , they are split into a fixed number of n windows in the spatial dimension, denoted as $X' \in \mathbb{R}^{n \times uhw \times c}$ and $Y' \in \mathbb{R}^{n \times hw \times C}$, where n represents the number of blocks, uhw and hw represent the window sizes of X' and Y' , and c and C represent the number of channels in X' and Y' , respectively. It is important to note that uhw and hw are related by powers of 2, and similarly, c and C are also related as powers of 2.

In a manner similar to the spatial information part, directly computing with the complete features would make it difficult for the model to focus on capturing channel-related information, while the substantial computational resources required pose another challenge. To address this, we first utilize rectangular strided convolutions to compress the windowed feature spatial dimensions while simultaneously converting the channel dimension of X' to match that of Y' . In this process, the model primarily focuses on the channel-wise information within the features and optimizes each channel based on the corresponding spatial information within the defined window, thereby enabling more efficient channel interactions and reducing interference from spatial noise. We denote the processed target features as Y_q , while the reference features are denoted as X_k and X_v , which contain the precise channel information lacking in the target features. To obtain precise channel information, we similarly input Y_q , X_k , and X_v to compute the required windowed channel information A and cross-channel information H along the channel dimension.

Subsequently, the obtained channel information H is input into a Feed-Forward Network (FFN) to enhance its representational capacity, and the final cross-channel information H is derived using the Sigmoid function. During this process, both intra-window and inter-window channel information are integrated into H to enhance the representation of channel information.

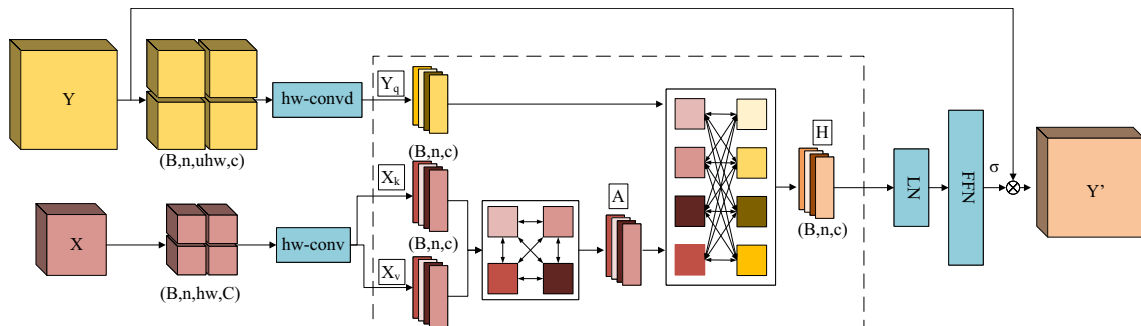


Fig. 7. Windowed channel information redistribution module (W-CIR).

Algorithm 3 Bottom-up cross-channel information flow (CCIF) with W-CIR.

Require: Target feature $x_i \in \mathbb{R}^{C \times H \times W}$, reference feature $d_f \in \mathbb{R}^{C_d \times H_d \times W_d}$

Ensure: Channel-refined feature y_{ch}

```

1: Divide  $x_i$  and  $d_f$  into  $n$  spatial windows
2: Flatten each window into tokens:  $x_\omega \in \mathbb{R}^{hw \times C}$ ,  $d_\omega \in \mathbb{R}^{hw \times C_d}$ 
3: for each window  $\omega = 1$  to  $n$  do
4:   Channel embedding:  $q_\omega \leftarrow \text{Conv1D}(x_\omega)$ ,  $k_\omega, v_\omega \leftarrow \text{Conv1D}(d_\omega)$ 
5:   Compute channel affinity:  $A_\omega = \text{Softmax}(q_\omega k_\omega^\top / \sqrt{d})$ 
6:   Aggregate channel context:  $H_\omega = A_\omega v_\omega$ 
7:   Apply feed-forward transformation:  $H'_\omega = \text{FFN}(H_\omega)$ 
8:   Channel gating:  $M_{ch,\omega} = \sigma(H'_\omega)$ 
9:   Reweight feature channels:  $y_{ch,\omega} = x_\omega \odot M_{ch,\omega}$ 
10: end for
11: Merge windows:  $y_{ch} \leftarrow \text{Concat}(\{y_{ch,\omega}\}_{\omega=1}^n)$ 
12: return  $y_{ch}$ 

```

Benefiting from the earlier compression of irrelevant spatial information, this structure enables the broad aggregation of channel context information within very limited computational resources and achieves targeted enhancement of channel information within specific windows through interactions between different windows. Finally, we apply the Hadamard product to the target features Y and the windowed cross-channel information H to obtain the channel-reallocated Y' .

Algorithm 3 outlines the overall CCIF process. The CCIF module conveys discriminative semantic cues from deeper layers to shallower ones using windowed channel information redistribution (W-CIR). It refines each feature's channel distribution by assigning adaptive importance weights to different channel groups, ensuring balanced and context-aware representation.

In this process, the shallow and deep features are aligned along the channel dimension, enabling them to establish comparability at a global statistical level. The shallow features capture the overall channel response pattern, while the deep features provide more discriminative channel representations. Through projection and similarity computation, a set of weighting coefficients reflecting channel importance is generated. Based on these coefficients, the channels of the deep features are reweighted to emphasize key channels consistent with the shallow features while suppressing irrelevant or redundant ones. Consequently, the shallow features, guided by the semantic information from the deep layers, acquire more accurate channel representations, whereas the deep features integrate the global statistical constraints provided by the shallow layers. Ultimately, the output of each network layer maintains a complete characterization of channel selection, ensuring that lesion regions are fully activated at the semantic level.

4. Experiments

4.1. Datasets and evaluation metrics

We conducted experiments on two commonly used retinal disease segmentation datasets, DDR [36] and IDRID [37], where the IDRID dataset contains fewer images but higher quality, while the DDR dataset includes a large number of images with varying quality.

The color fundus images in the DDR dataset were collected from 147 hospitals across 23 provinces in China and consist of a total of 13,673 retinal images, with 757 images providing pixel-level annotations needed for segmentation, including EX, HE, MA, and SE with 486, 610, 570, and 239 pixel-level annotated images, respectively. These 757 images were divided into 383 for training, 149 for validation, and 225 for testing, and our experiments followed this division.

The IDRID dataset consists of images collected from a single eye hospital in Mumbai, India, all of which were captured using the same digital fundus camera. The dataset includes 81 images for retinal disease

segmentation, with the same EX, HE, MA, and SE classes, with EX and MA having 81 pixel-level annotations each, and HE and SE having 80 and 40 pixel-level annotations, respectively. The 81 images were divided into 54 for training and 27 for testing, with our experiments following this division.

In our experiments, we used the most commonly used IoU, Dice, and AUPR as evaluation metrics, including class-specific and average metrics. IoU is the most widely used evaluation metric for natural image segmentation tasks, Dice is the most widely used evaluation metric for medical image segmentation tasks, and AUPR is the most effective evaluation metric for evaluating the usefulness of lesion segmentation.

4.2. Implementation details

Our network was built in a Python 3.8 environment using the PyTorch 2.0 framework, and all experiments were conducted on an RTX 3090 GPU. We used the MMSegmentation [38] official deep learning program framework for model building.

On the DDR dataset, we resized images based on the Dual-PSPNet [39] and L-seg [31] papers, i.e., we used equal proportions to make the long edge 1024 pixels and then used zero padding to increase the short edge to 1024 pixels. On the IDRID dataset, we resized the images, reducing them to 1440×960 pixels. Simultaneously, we performed the same data augmentation methods, including multi-scaling (0.5–2.0), rotation (90, 180, 270), and flipping, on the data.

On all datasets, we used the SGD optimizer to train the model. We employ the commonly used Cross-Entropy Loss (CE-Loss) and Dice Loss, combining them in a 1:1 ratio to form the training loss function. This combination mitigates the negative impact of class imbalance during segmentation tasks. The initial learning rate was 0.01, momentum was fixed at 0.9, weight decay was 0.0005, and the learning rate was decayed using the power = 2 PolyLR. The batch size was fixed at 2 during training and 1 during testing to approximate clinical usage scenarios.

4.3. Ablation study

An ablation study is conducted to analyze the individual contributions of each IFR component. This analysis reveals how spatial and channel information flows collaboratively improve segmentation accuracy. We conducted ablation studies on the DDR and IDRID datasets. In the ablation studies, we followed the recommendations of the dataset [36,37] and focused on comparing the AUC_PR and average AUC_PR for each class.

Table 1 demonstrates that the integration of CSIF significantly enhances the overall segmentation performance of the model on the DDR dataset, with particularly notable improvements in the EX and SE categories, increasing by 2.25% and 3.23%, respectively. Similarly, incorporating CCIF also leads to performance gains across all four lesion types. Compared to CSIF, CCIF shows greater improvements in HE and MA segmentation, with increases of 5.37% and 0.7%, respectively. In terms of overall segmentation capability, CCIF alone contributes more effectively than CSIF alone. This can be attributed to HRNet's high-resolution convolution method, which preserves more information in the channel features. When both CSIF and CCIF are applied together, the model achieves a significant boost in overall segmentation performance. Although there is a slight decrease in EX performance compared to using only CSIF or CCIF, substantial improvements are observed in the more challenging categories of HE, SE, and MA. This indicates that the

Table 1
Ablation study on the DDR dataset.

SIR	CIR	mAUPR	EX	HE	SE	MA
		47.26	62.72	50.7	49.32	26.32
✓		48.87	64.97	52.97	51.55	26.01
	✓	49.59	64.21	56.07	51.02	27.05
✓	✓	50.96	64.00	55.55	56.45	27.82

Table 2
Ablation study on the IDRid dataset.

SIR	CIR	mAUPR	EX	HE	SE	MA
		65.24	82.66	64.13	69.25	44.94
✓		66.78	81.88	69.05	73.87	42.33
	✓	68.12	82.35	68.27	74.93	46.91
✓	✓	68.99	83.52	68.37	74.33	49.75

model places greater emphasis on difficult-to-segment lesions, achieving more balanced and effective overall segmentation performance.

Table 2 shows that the introduction of CSIF on the IDRid dataset leads to substantial improvements in the segmentation of HE and SE, with increases of 4.92% and 4.62%, respectively. Since microaneurysms (MA) occupy only a very small portion of the spatial domain in retinal images, introducing a spatial information refinement module alone has limited positive impact. After incorporating CCIF, the model exhibits notable performance gains, particularly in the SE and MA categories, improving by 5.68% and 1.97%, respectively. In terms of overall segmentation capability, the performance improvement from using CCIF alone surpasses that from CSIF alone, which aligns with the results observed on the DDR dataset. This suggests that HRNet’s high-resolution convolutional learning is more effective at leveraging channel-wise information. When both CSIF and CCIF are utilized simultaneously, the model achieves further improvements across all categories, with gains of 0.86%, 4.24%, 5.08%, and 4.81% in EX, HE, SE, and MA, respectively. The significantly greater enhancements in HE, SE, and MA compared to EX confirm that the proposed method strengthens the model’s ability to focus on more challenging lesion types.

When both CSIF and CCIF are used, the model’s overall segmentation performance improves significantly, with improvements of 0.86, 4.24, 5.08, and 4.81 in the EX, HE, SE, and MA classes, respectively. The improvements in the HE, SE, and MA classes are much more pronounced than in the EX class, further indicating that the model pays more attention to the more difficult-to-segment classes.

4.4. Comparative study

4.4.1. DDR

We compared our proposed IFR method with several state-of-the-art approaches for retinal lesion segmentation on the DDR dataset. The evaluation focused on three commonly used metrics in the field of retinal disease segmentation: AUPR, IoU, and Dice coefficient. Table 3 presents the comparative results on the DDR dataset. Given that all methods used the same dataset split and image resolution, the performance of the referenced methods is cited directly from their original publications.

However, due to the complexity and challenges inherent in retinal lesion segmentation, there are relatively few existing domain-specific

methods. To enrich the comparison, we also reproduced several representative advanced methods from the general segmentation domain using the MMsegmentation framework. All reported results are averaged over three independent runs to ensure robustness and reliability.

As shown in Table 3, the proposed method (OUR) consistently outperforms existing state-of-the-art approaches in retinal disease segmentation on the DDR test set. Compared to all other methods, OUR achieves the highest mean AUPR (50.96), mean IoU (31.27), and mean Dice score (46.73), indicating strong and consistent overall performance. The improvement is particularly pronounced in soft exudates (SE), the lesion types that are small, sparsely distributed, and low in contrast. This suggests that IFR effectively enhances both boundary delineation and lesion completeness through the balanced redistribution of spatial and channel information. While methods such as M2MRF-C and HRNetV2 also show competitive results, OUR surpasses them notably in both lesion-wise and aggregate metrics, especially in SE and MA lesion categories where precision is challenging. Importantly, OUR outperforms HRNetV2* and CCNet*, both implemented under the same MMsegmentation framework, highlighting that the improvement is not due to implementation discrepancies but due to the inherent advantage of OUR’s architecture. This consistent advantage across all evaluation metrics and lesion types reflects OUR’s robustness, strong generalization ability, and practical value in real-world medical image segmentation tasks.

The reason for this result is that EX and SE belong to the same bright yellow lesions, and existing methods are often biased towards estimating EX and ignoring SE. Our spatial and channel information flow can complete the information of different layer features, allowing the model to distinguish EX and SE, well and yield more balanced segmentation results. In Fig. 8, we visualized the segmentation results and performed a qualitative comparison. Compared to other methods, our approach achieves the best results for both small and sparse lesions, as well as for lesions with large size distributions.

From Fig. 8, we can observe that CCNet can only detect the most prominent lesions, and is not able to detect lesions with small color differences. Compared to these methods, M2MRF’s recognition performance is better, but it seems to focus more on local color information and ignores the overall spatial and channel correlations, which causes it to misclassify normal parts of the retina as SE lesions and classify regions with low contrast as normal.

Compared to other methods, our method’s segmentation structure is most similar to the actual disease situation. Thanks to the window-based spatial and channel information flow, which can reasonably distribute the spatial and channel information contained in different features, allowing our method to achieve more accurate segmentation of irregular lesion areas. As seen in the fourth row, our method can effectively

Table 3

Segmentation results on the DDR test set, “*” denotes the results obtained by our implementation, “-” indicates that experimental data were not provided in the original literature.

Methods	EX			HE			SE			MA			Mean		
	AUPR	IoU	Dice	AUPR	IoU	Dice	AUPR	IoU	Dice	AUPR	IoU	Dice	AUPR	IoU	Dice
L-seg Ref. [31]	55.46	-	-	35.86	-	-	26.48	-	-	10.52	-	-	32.08	-	-
Dual-PSPNet Ref. [39]	54.24	39.85	56.99	-	-	-	-	-	-	-	-	-	-	-	-
DeepLabV3+ Ref. [29]	-	-	-	-	-	-	-	-	-	-	-	-	35.60	25.76	39.69
Dual-Branch Ref. [26]	58.44	-	56.47	49.52	-	49.91	23.45	-	21.22	19.18	-	28.28	37.64	-	38.97
Trans2U-Net Ref. [27]	59.79	-	-	48.07	-	-	42.43	-	-	18.23	-	-	42.13	-	-
M2MRF-C Ref. [32]	63.59	43.49	60.62	54.43	29.17	45.16	49.35	31.39	47.78	28.38	16.31	28.04	48.94	30.09	45.
VTA + HBE Ref. [40]	56.07	-	-	37.78	-	-	26.94	-	-	10.66	-	-	-	-	-
Cosed-Net Ref. [41]	-	39.87	57.01	-	24.1	38.84	-	20.88	34.54	-	10.34	18.74	-	23.80	-
HRNetV2* Ref. [42]	62.72	42.04	59.20	50.70	27.33	42.92	49.32	29.28	45.30	26.32	15.09	26.22	47.26	28.43	43.41
CCNet* Ref. [13]	55.65	18.87	31.75	32.29	9.67	17.63	37.07	24.41	39.24	5.48	6.99	13.07	32.62	14.99	25.42
SEBNet Ref. [46]	65.30	35.77	52.69	46.31	28.53	44.40	3.72	28.50	44.36	15.78	8.72	16.03	42.78	25.38	39.37
OUR	64.00	41.53	58.69	55.55	28.73	44.64	56.45	38.75	55.85	27.82	16.09	27.74	50.96	31.27	46.73

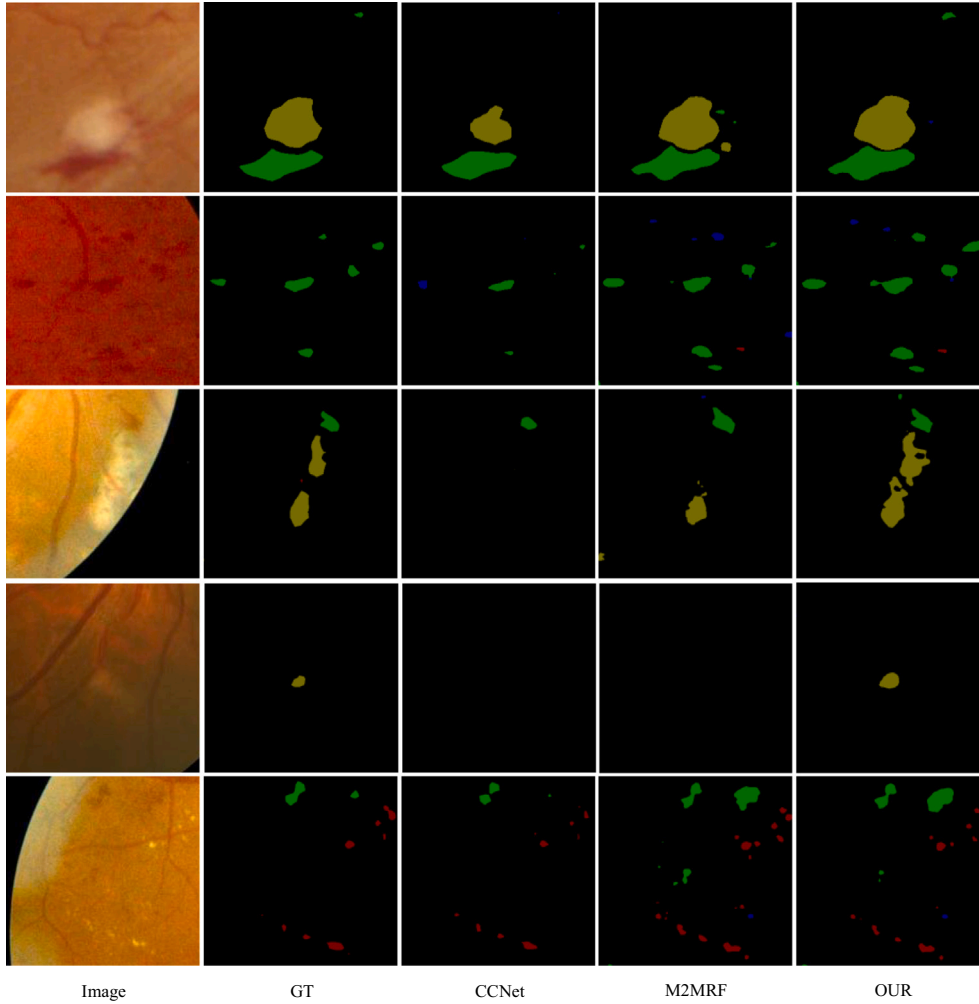


Fig. 8. Qualitative comparison. The first column is the original input image, the second column is ground truth.

identify lesions with blurred boundaries that are similar to the background information. The second row shows that our method maximizes detection of small and scattered lesions, effectively identifying all lesion regions. IFR achieves more accurate segmentation for small and blurred lesions. Enlarged regions highlight the model's strong ability to capture microaneurysms and soft exudates with clear structural boundaries and reduced background noise. This advantage stems from the top-down CSIF flow, which transfers detailed spatial cues to deeper layers, ensuring fine-grained localization even under weak contrast conditions.

4.4.2. IDRID

Similar to the evaluation conducted on the DDR dataset, we also compared the proposed IFR method with state-of-the-art approaches for retinal lesion segmentation on the IDRID dataset. Table 4 presents the comparison results on the IDRID dataset.

According to the segmentation results on the IDRID dataset shown in Table 4, OUR achieves the highest scores in mean IoU (51.45) and mean Dice (67.04), confirming its strong overall segmentation performance for retinal lesions. Although the mean AUPR (68.99) of OUR is slightly lower than that of SEBNet (71.01), OUR presents a more balanced and consistent performance across all four lesion types—EX, HE, SE, and MA—without significant weaknesses in any particular category. Especially in SE, a lesion type often considered difficult to segment due to its subtle appearance and similarity to other regions, OUR achieves notable improvement with an IoU of 56.93 and Dice of

72.56, outperforming all other methods in this category. This advantage arises from OUR's spatial and channel information reallocation, which enhances feature distinction and localization. Compared with strong competitors like M2MRF-C and HRNetV2, OUR shows better generalization and robustness across both bright and dark lesions. These findings suggest that OUR is not only highly accurate in individual lesion segmentation but also offers greater reliability for practical use in real-world diagnostic settings.

Quantitatively, IFR achieves the highest or comparable performance on both DDR and IDRID datasets. The consistent improvement across different lesion types demonstrates the robustness of our dual information flow design. Specifically, CSIF mitigates the loss of structural information caused by repeated downsampling, while CCIF strengthens shallow feature discrimination through channel refinement. This synergy allows IFR to maintain balanced accuracy across lesion categories, improving sensitivity for micro-lesions and stability for large-scale structures.

Some retinal lesions are extremely small and sparsely distributed, while others have significantly larger sizes. This vast difference in lesion size necessitates that features of different scales within the network contain sufficient and distinguishable spatial and channel information. To intuitively demonstrate the effectiveness of our method, we visualized the segmentation results in Fig. 9 and conducted qualitative comparisons. Compared to other methods, our approach achieves the best results for both small and sparse lesions, as well as for lesions with large size distributions.

Table 4

Segmentation results on the IDRID test set, “*” denotes the results obtained by our implementation, “-” indicates that experimental data were not provided in the original literature.

Methods	EX			HE			SE			MA			Mean		
	AUPR	IoU	Dice	AUPR	IoU	Dice	AUPR	IoU	Dice	AUPR	IoU	Dice	AUPR	IoU	Dice
VRT(1st) Ref. [43]	71.27	-	-	68.04	-	-	69.95	-	-	49.51	-	-	64.69	-	-
L-seg Ref. [31]	79.45	-	-	63.74	-	-	71.13	-	-	46.27	-	-	65.15	-	-
Dual-PSPNet Ref. [39]	77.48	60.95	75.74	-	-	-	-	-	-	-	-	-	-	-	-
DeepLabV3+ Ref. [29]	-	-	-	-	-	-	-	-	-	-	-	-	59.08	38.82	54.13
Simultaneous Ref. [28]	79.58	-	76.32	69.14	-	66.54	72.67	-	49.88	38.04	-	36.02	65.86	-	57.19
Trans2U-Net Ref. [27]	89.25	-	-	68.91	-	-	62.31	-	-	49.29	-	-	67.44	-	-
VTA + HBE Ref. [40]	86.40	-	-	64.94	-	-	59.78	-	-	40.63	-	-	62.94	-	-
nmODE-Unet Ref. [44]	83.61	64.24	75.15	62.62	44.73	60.37	73.81	53.72	69.81	51.77	36.17	52.62	67.95	49.72	64.49
M2MRF-C Ref. [32]	82.16	66.46	79.85	68.69	49.72	66.42	69.32	51.43	67.92	48.80	32.13	48.63	67.24	49.94	65.71
HRNetV2* Ref. [32]	82.66	65.63	79.25	64.13	42.85	59.94	69.25	50.12	66.59	44.94	28.34	44.15	65.24	46.73	62.48
Ddrnet* Ref. [45]	62.52	3.77	7.26	47.31	1.67	3.29	67.86	23.16	37.97	20.69	1.60	3.15	49.59	7.62	12.91
CCnet* Ref. [13]	67.31	17.86	30.30	62.89	20.79	34.42	68.26	35.10	51.96	36.66	5.71	10.81	58.78	19.86	31.87
SEBNet Ref. [46]	88.36	66.45	79.84	69.71	44.80	61.87	76.69	56.11	71.89	49.26	21.89	35.92	71.01	47.31	62.38
OUR	83.52	66.72	80.04	68.37	48.92	65.70	74.33	56.93	72.56	49.75	33.21	49.86	68.99	51.45	67.04

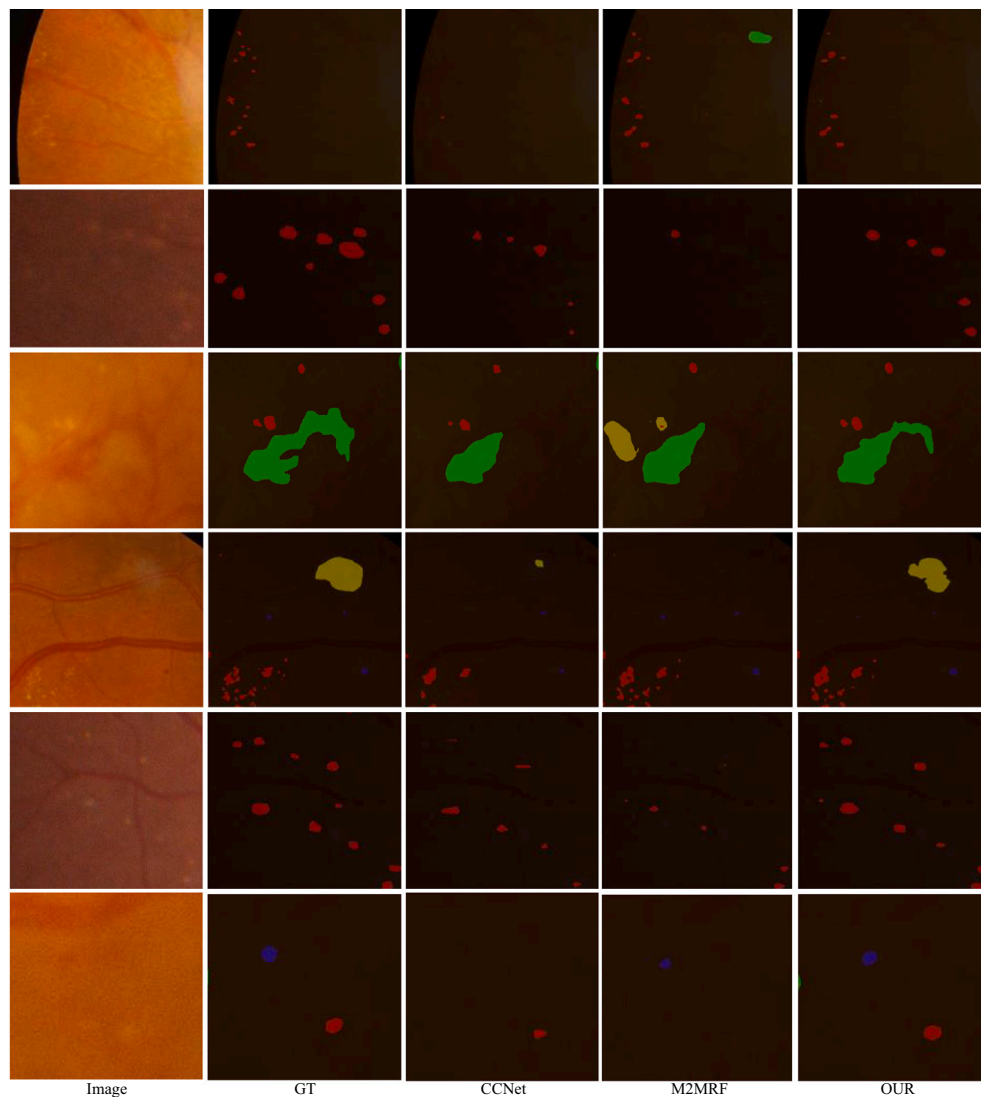


Fig. 9. Qualitative comparison. The first column is the original input image, the second column is ground truth.

From Fig. 9 we can observe that CCNet only detects the most prominent lesions, and is not able to detect lesions with small color differences. M2MRF seems to focus more on local color information and ignores the

overall spatial and channel correlations, which causes it to misclassify normal parts of the retina as SE lesions and classify regions with low contrast as normal.

Table 5

Cross-domain generalization study. All models were trained on the training set of the DDR dataset and tested on the IDRID dataset's test set.

Methods	AUPR				
	mAUPR	EX	HE	SE	MA
L-seg Ref. [31]	39.88	65.01	44.05	30.59	19.86
RTNet Ref. [30]	42.05	67.99	45.04	34.01	21.14
HRNetV2 Ref. [28]	56.33	68.71	60.12	54.83	41.66
M2MRF-C Ref. [32]	59.3	73.94	61.01	57.61	44.64
OUR	61.05	75.31	63.43	64.32	41.12

Compared to other methods, our method's segmentation structure is most similar to the actual disease situation. As shown in the third row of Fig. 9, the green area (HE) benefits significantly from the windowed spatial and channel information flows. These mechanisms enable the reasonable redistribution of spatial and channel information across different features, allowing our method to achieve more accurate segmentation of particularly irregular lesion regions. In Fig. 9, the first and second rows further illustrate that our method outperforms other methods in detecting small and scattered lesions. The fourth row demonstrates that our approach can also effectively detect lesions with blurred boundaries and significant size and distribution differences.

The results show that the proposed IFR method not only preserves the integrity of small lesions but also reconstructs blurred lesion boundaries more precisely. This demonstrates that the CSIF and CCIF modules complement each other—CSIF strengthens spatial sensitivity by propagating fine positional details downward, while CCIF enhances channel discriminability by guiding shallow layers with deeper semantic cues. Together, they allow the network to effectively distinguish subtle pathological areas from surrounding tissues.

4.5. Generalization study

To validate the model's generalization performance, we trained the model using the training set of the DDR dataset and tested it on the IDRID dataset's test set. The L-seg and RTNet come from [32], HRNetV2 and M2MRF-C come from [37].

From Table 5 we can see that our method achieved the best average performance, increasing by 1.75 compared to the second place, indicating that our method has better cross-domain generalization ability. Similarly, we achieved the best performance in the EX, HE, and SE classes, especially in the SE class, increasing by 6.71 compared to the second place, indicating that our model can still distinguish easily confusing bright yellow lesions even in different data domains.

5. Conclusion and future work

In this paper, we propose an information flow redistribution (IFR) framework from the perspective of spatial and channel information flows. Since deep features usually possess more accurate channel information but lack fine spatial details, while shallow features show the opposite pattern, we designed a bidirectional mechanism—comprising the top-down Cross-Spatial Information Flow (CSIF) and the bottom-up Cross-Channel Information Flow (CCIF)—to explicitly exchange and reassign complementary information across feature hierarchies. This ensures that each feature layer retains sufficient spatial and channel cues to effectively activate lesion-related regions and achieve a more balanced feature representation.

Through comprehensive qualitative and quantitative experiments, the proposed IFR demonstrated reliable and consistent performance in retinal lesion segmentation. It effectively alleviates boundary blurring and adapts to lesions of different sizes and distributions. The redistributed features enhance the network's ability to separate closely clustered pathological regions, resulting in more coherent segmentation outputs. Moreover, the framework exhibits strong cross-domain

generalization across various imaging conditions and acquisition devices, suggesting its robustness and potential for practical deployment in clinical workflows.

Beyond diabetic retinopathy, the proposed IFR framework is not restricted to retinal analysis but provides a general strategy for medical image segmentation. Its modular and architecture-agnostic design allows straightforward integration into diverse segmentation networks. Potential applications include 2D tasks such as optic disc/cup and vessel segmentation, dermoscopic lesion analysis, and cell instance segmentation, as well as 3D volumetric tasks such as tumor delineation in CT/MRI and multi-organ segmentation. The information flow redistribution mechanism can also support multi-modal fusion scenarios by harmonizing spatial and channel information across different feature domains. Although IFR is designed for offline medical image segmentation rather than real-time control systems, its training stability also benefits from principles commonly discussed in data-driven modeling. The diversity of retinal images provides sufficiently rich input variations, which facilitates stable optimization of the redistribution modules. In addition, regularization mechanisms such as dropout and numerical stabilization help maintain reliable feature learning without increasing computational burden, ensuring robustness across heterogeneous data domains.

For future work, we plan to (1) explore more efficient implementations of the windowed redistribution modules to reduce computational costs while maintaining accuracy; (2) integrate the IFR paradigm with hybrid CNN–Transformer architectures to combine explicit cross-layer redistribution with global contextual reasoning; and (3) extend validation to additional clinical modalities, including volumetric and multi-modal datasets, to further assess generalization and adaptability. We believe that the proposed IFR framework offers a promising and versatile foundation for improving spatial–channel representation balance across a wide range of medical image analysis tasks.

CRedit authorship contribution statement

Jiale Wang: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Hui Ma:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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